Bryson Loflin
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EDUCATION

Ph.D., Ecology and Evolutionary Biology, Princeton University, Princeton, NJ 2025 - Present
B.S., Department of Biology, UNC Chapel Hill, Chapel Hill, NC 2021 - 2025

RESEARCH INTERESTS

Combining field and laboratory work while using animal systems, with a focus on cooperative social behaviors, to answer questions in ecology and evolution.

RESEARCH EXPERIENCE

Graduate Research
Dr. Gerald Carter, Associate Professor, Department of Ecology and Evolutionary Biology, Princeton
Projects:

- Reciprocity and fitness interdependence in vampire bats — Summer 2025 - Present
- Bayesian social relations models of grooming and food sharing — 2025 - Present
- Defection experiments to test partner choice in captive colonies — 2026 - Present
- Machine learning to identify individual bats from video — 2026 - Present

Undergraduate Research
Dr. Karin Pfennig, Professor, Department of Biology, UNC Chapel Hill
Projects:

- Evaluated presence and distribution of a behavioral syndrome — 2022 - 2025
- Genotyped individuals to evaluate species co occurrence — Spring 2021
- Evaluated adaptive hybridization through a literature review — Fall 2020

Undergraduate Research
Dr. Michael Sheehan, Associate Professor, Department of Neurobiology and Behavior, Cornell
Project:

- Quantify the behavioral impacts of mouse rewilding — Summer 2024

TEACHING EXPERIENCE

Teaching Assistant; Life on Earth Fall 2025
Department of Ecology and Evolutionary Biology, Princeton

Teaching Assistant; Seafood Forensics Fall 2023 - Spring 2025
Department of Biology, UNC Chapel Hill

Teaching Assistant; Investigating the Biodiversity Crisis Spring 2024
Department of Biology, UNC Chapel Hill

WORK EXPERIENCE

Senior Residential Computing Consultant Fall 2023 - Spring 2025
Carolina Housing, UNC Chapel Hill

Residential Computing Consultant Fall 2021 - Spring 2022
ResNET, Carolina Housing, UNC Chapel Hill

PUBLICATIONS

Zipple, M. N., **Loflin, B.**, Chang Kuo, D., Tan, E., & Sheehan, M. J. (2025). Transfer to a naturalistic setting restructures fear responses in laboratory mice. *Current Biology*, 35(24), R1175-R1176. https://doi.org/10.1016/j.cub.2025.10.050

Ryan, S., Kelly, P., **Loflin, B.**, & Pfennig, K. S. (2025). Experimentally testing the hypothesis that hybrid fitness can increase evolutionarily: An example in spadefoot toads. *Ecology and Evolution*, 15(12), e72531. https://doi.org/10.1002/ece3.72531

Ryburn, S. J., Yu, T., Ong, K. J., Wisely, E., Alston, M. A., Howie, E., LeRoy, P., Giang, S. E., Ball, W., Benton, J., Calhoun, R., Favreau, I., Gutierrez, A., Hallac, K., Hanson, D., Hibbard, T., **Loflin, B.**, Lopez, J., Mock, G., … Bruno, J. F. (2025). Sale of critically endangered sharks in the United States. *Frontiers in Marine Science*, 12, 1604454. https://doi.org/10.3389/fmars.2025.1604454

TALKS AND PRESENTATIONS

ABS Poster: *Cooperative relationship formation in vampire bats: Does helping become more symmetrical over time?* Summer 2026
Senior Honors Thesis Presentation Spring 2025
UNC Research and Discovery Fair Fall 2024
ABS Co-Author: Behavior's role in the evolution of species ranges Summer 2024
Research Symposium: *A behavioral syndrome in Spea bombifrons* Spring 2022
Research Symposium: *The sale of critically endangered sharks in the United States* Spring 2022

HARD SKILLS

- Bayesian social relations models and network analysis in R (STRAND, Stan, brms)
- Machine learning re-identification of individual animals from video (Python, SLEAP)
- Data analysis in R and JMP
- Captive vampire bat husbandry and fasting trials
- Scoring social behavior from infrared video (BORIS)
- RFID chipping of mice and RFID tracking of bats
- Open field and mate choice tests on spadefoot toads
- Plus maze tests, submandibular blood collection, and capture of rewilded mice
- Capture of spadefoot toads and transport of animals between lab and field sites
- Novel protocol development for behavioral assays
- DNA extraction, PCR, and preparation of samples for Sanger Sequencing
- Technology troubleshooting

SOFT SKILLS

- Communicating data
- Team management
- Time management
- Mentee training
- Academic writing

HONORS AND AWARDS

NSF Graduate Research Fellowship Program Honorable Mention Spring 2025, 2026
- Awarded by the National Science Foundation to highly ranked applicants

Centennial Fellowship in the Natural Sciences and Engineering Fall 2025
- Awarded by the Princeton University Graduate School

Robert Coker Award Spring 2025
- Given to one senior biology major each year for excellence in research

Highest Honors Senior Thesis Spring 2025
- Awarded for meeting the most rigorous standards of scholarly excellence

Carolina Housing Hall of Fame Spring 2024
- Awarded to Carolina Housing staff for exceptional service

RCC of the Semester Spring 2022
- Awarded to one of the 40+ staff team for contributions to ResNET

North Carolina Association of Student Councils Hall of Fame Summer 2018
- Awarded to individuals for distinguished service towards their student council

OUTREACH AND SERVICE

- Out in STEM (oSTEM)
- Undergraduate research mentoring
- Vampire bat cooperation game and science pages at brysonloflin.com

REFERENCES

Gerald Carter, Associate Professor
Department of Ecology and Evolutionary Biology
Princeton University
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Karin Pfennig, Professor
Department of Biology
University of North Carolina at Chapel Hill
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Michael Sheehan, Associate Professor
Department of Neurobiology and Behavior
Cornell University
msheehan@cornell.edu

John Bruno, Chi Omega Distinguished Professor
Department of Biology
University of North Carolina at Chapel Hill
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